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PHY 415 — Past Questions & Materials
by Renance

Course: PHY 415
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 2
Levels: 400

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Paper 1 of 2 — PHY 415 Mathematics Methods for Physics I

Level 400 • Second Semester • PastQuestion

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COS/6085/2019

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
 COLLEGE OF SCIENCE
 DEPARTMENT OF PHYSICS
 SECOND SEMESTER EXAMINATION

SESSION: 2023/2024 Academic Session
 COURSE CODE: PHY 415
 COURSE TITLE: Mathematical Methods for Physics I.
 TIME ALLOWED: 3 hrs.
 COURSE UNIT: 3.

Instructions: You are expected to answer 4 out of the available 6 questions. All questions carry equal marks, and all the symbols retain their usual meaning.

Question 1. (a) State the Chebyshev polynomial of the first and second kind. (b) Show that the Chebyshev polynomial of the first kind and the second kind, are independent solutions of the Chebyshev differential equation

$$(1-x^2)\frac{d^2y}{dx^2} - x\frac{dy}{dx} + n^2y = 0$$

(c) State the recurrence relation of Chebyshev polynomial and solve for: T_1 , T_2 , T_3 , T_4 , T_5 and T_6 .

Question 2. Use matrices to solve the following homogeneous differential equations:

(a)
$$\frac{d^2x}{dt^2} - 5\frac{dx}{dt} + 6x = 0$$

subject to the initial boundary condition: $x(0) = 1$, $x'(0) = 2$

(b)
$$\frac{d^2y}{dx^2} + 4y = 0$$

subject to the initial values: $y(0) = 1$, $y'(0) = 0$.

Question 3. Find the eigenvalues and eigenvectors of the following regular Sturm-Liouville problem:

(a) $\phi'' + \lambda\phi = 0$

subject to the Boundary Conditions BCs: $\phi(0) = \phi(L) = 0$, where L is some positive number.

(b) $y'' + \lambda y = 0$; $0 < x < L$

subject to the Boundary Conditions BCs: $y(0) = 0$, $y(L) + y'(L) = 0$ where L is some positive number.

Question 4. Define the Gamma function and evaluate the following Gamma functions:

$$(a) \int_0^{\infty} \sqrt{x} e^{-\sqrt[3]{x}} dx \quad (b) \int_0^{\infty} e^{-h^2 x^2} dx$$

(c). Define the Beta function and use it to evaluate Beta function:

$$\int_0^1 x^4 (1-\sqrt{x})^5 dx.$$

Question 5. (a) Solve the non-homogeneous equation

$$y'' + 4y = 8x^2$$

(b). Solve the following boundary value problem

$$y'' + 9y = \cos x$$

subject to the boundary value condition: $y'(0) = 5$, $y(\pi/2) = -\frac{5}{3}$.

Question 6. (a) Prove that $u = x^2 - y^2$ and $v = \frac{y}{x^2 + y^2}$ are harmonic functions of (x, y) , but are not harmonic conjugates.

(b) If $u - v = (x - y)(x^2 - 4xy + y^2)$ and $f(z) = u + iv$ is an analytic function of $z = x + iy$, find $f(z)$ in terms of z .

(c) (i) State Cauchy-Riemann theorem, and show that the function

$U = e^{-x}(x \sin y - y \cos y)$ is harmonic (ii) find the function $V(x, y)$ such that $f(z) =$

$U + iV$ is analytic.

Paper 2 of 2 — Mathematical methods for physics I PHY 415 PQ

Level 400 • First Semester • PastQuestion

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